When Is Control-Aware Communication Locally Decidable? Information-Structure Limits in Distributed Formation Control

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***Abstract*—Control-aware communication assigns network resources according to the closed-loop consequence of an update. This premise hides an information question: does the transmitting agent possess enough information to determine that consequence? We study finite-dimensional sampled linear systems for which a communication action adds a fixed response to a finite-horizon performance output. The exact quadratic action benefit is an affine functional of an augmented closed-loop state. Building on functional observability, we characterize exact value identifiability from current and finite-history agent information by row-space conditions. We then quantify the irreducible value-estimation radius and the strictly positive local minimax communication-decision regret when compatible values straddle zero. The minimum number of supplementary linear scalar statistics needed for several action values equals the rank of their unobservable coefficient projection. Coalition ownership of those statistics and their communication acquisition cost are treated separately.**

The framework is instantiated on an implementation-derived, sampled five-UAV formation controller with delayed and lossy communication. All ten controller-relevant action values violate sender-history identifiability in the evaluated information topology. Two full, unsaturated, dynamically reachable histories—one ordinary payload and one pinned-leader payload—produce identical allowed sender observations and identical action responses but opposite exact action-value signs. Although one scalar closes each fixed-action algebraic deficit, exhaustive coalition analysis requires all five agents to synthesize it in this topology. Across each sender's candidate set, four leader actions share three hidden directions, whereas the other multi-action senders show no compression. Frozen development experiments further show why this distinction matters: centralized exact value has scheduling headroom in a formation-switching regime, whereas charging for the information needed to obtain that value can remove the headroom. The results delimit exact local value-based communication; they do not rule out prior-specific Bayesian decisions or nonlinear policies.

***Index Terms*—Networked control systems, distributed control, information structures, functional observability, value of information, event-triggered control, multi-agent formation.**

I. Introduction

Communication policies for networked control are increasingly asked to send the packet that matters most to the control objective, rather than simply the oldest packet or the packet with the largest state error. Event-triggered control formalizes state- or error-dependent transmissions [1]–[3]; age of information (AoI) formalizes freshness [4], [5]; and value-of-information (VoI) formulations attach a control consequence to an update [6]. These approaches answer different questions. AoI describes how stale a record is, while a finite-horizon control value asks how the closed loop will change if that record is refreshed.

The latter question is meaningful only relative to an information structure. In a distributed formation, the sender knows its own physical and controller state and its transmission history. It need not know the packet currently held by a receiver, the receiver's other stale memories, or remote states that shape the no-action closed-loop trajectory. Acknowledgments can reduce one part of this uncertainty, and ACK-free packet-memory beliefs can be calibrated under a known channel law. Neither fact implies that the total control-action value is locally computable: the cross term between the action response and the global no-action response may remain hidden.

This paper therefore asks a question prior to scheduler design:

> Under what information structures is the exact control value of a communication action identifiable, and what additional information must be acquired when it is not?

The question connects decentralized information structures [7]–[9] with functional observability [10], [11]. We do not claim the row-space test itself as new. Rather, we expose the exact communication-action value as the target functional, show that inability to observe its magnitude can extend to its sign, separate the dimension of the missing statistic from the network cost of constructing it, and close these statements on an implemented formation-control model.

The investigation was driven by falsification. A deterministic communication-error bound was analytically valid but produced a Pareto frontier dominated by periodic communication. An online centralized current-state oracle, using the exact action cross term but no future realizations, exposed substantial scheduling headroom in some nonstationary regimes and none in a congestion boundary. Explicit receiver feedback often consumed that headroom. Finally, an exact ACK-free receiver-memory belief was well calibrated but could not identify the baseline component of the action cross term. These findings motivated an information-limit result rather than another trigger.

The contributions are:

1) We derive the exact finite-horizon marginal

communication-action value as an affine functional of an augmented state for a fixed action response.

2) Using functional-observability geometry, we quantify the irreducible value-information radius and derive deterministic and randomized local minimax communication-decision regret when compatible exact values straddle zero.
3) We prove that the minimum dimension of extra instantaneous linear information for a family of actions is the rank of its hidden projection, and apply it sender-wise to expose shared versus independent missing directions. We separately characterize coalition ownership because statistic dimension is not packet count or distributed acquisition cost.
4) For the sampled N=5 formation architecture, we validate the affine map and rank conclusions using tolerance sweeps and 80-digit arithmetic, show sender-history nonidentifiability for all ten payloads, construct two reachable opposite-sign histories with positive unavoidable regret, and connect the missing information to a measured feedback break-even condition.

The results are scoped to exact affine action-value identifiability in the stated sampled linear model. They do not assert that distributed agents can never make useful stochastic decisions, that feedback is always uneconomic, or that adaptive communication is universally inferior to periodic service.

II. Related Work and Positioning

A. From event conditions to stale receiver memories

Event-triggered control replaces periodic sampling by state-dependent conditions while retaining stability or performance guarantees [1]–[3], [12], [13]. Loss and delay make the sender and receiver disagree about the control information in use. Decentralized double-integrator consensus under unreliable links is treated in [14]; ACK and no-ACK event-triggered designs under packet loss appear in [15]; and model-based triggering without ACK appears in [16]. Most closely, Viel et al. maintain packet-loss hypotheses for neighbor-held states in distributed formation control [17]. Thus no-ACK receiver memory inference is established prior art. Our issue is what remains hidden after that receiver-memory uncertainty is represented exactly.

B. Freshness and control value

AoI measures time since generation of the freshest received update [4]. Goal-oriented variants such as age of incorrect information refine freshness by state relevance [5], [18]. These metrics are useful network summaries but are not generally sufficient statistics for the finite-horizon marginal value of an action. VoI-based feedback control makes that marginal value explicit and can yield optimal threshold structures under its prescribed information model [6]. Formation scheduling has also combined value proxies, graph relevance, and resource allocation [19]. We neither introduce the VoI concept nor optimize a scheduler. We test whether the exact value required by such a decision is observable to the deciding agent.

C. Information structures and functional observability

Who knows what and when is central to decentralized control [7], [8]. Common-information methods can convert specific partial-history-sharing problems into coordinated stochastic control problems [9], but they require a declared probability model and information sharing architecture. Functional observability asks whether selected linear functionals, rather than the full state, can be reconstructed [10]. Its row-space characterizations and target-estimation questions are established, including for large networks [11]. Recent work provides modal and structural characterizations and optimal sensor-placement results [20], sample-based conditions for intermittent or irregular measurements [21], and reduced-order functional-observer existence and pole-placement methods [22]. Structural refinements cover generically diagonalizable systems [23] and fixed-observable-set characterizations [24]; target control/observation duality is developed in [25].

These results imply three explicit boundaries. Our current- and finite-history row-space conditions are applications of functional and sample-based functional observability, not new generic tests. Our coalition search is an instantiated functional sensor-selection problem, not a new optimization class. Finally, the instantaneous missing-statistic dimension derived here is neither minimum sensor count nor functional-observer order. Our addition is the communication-decision consequence: an exact action-value functional, its local minimax estimation and binary-decision regret, and the separation between simultaneous-action statistic dimension and network acquisition cost.

D. Control–communication co-design

Data-rate theorems and networked-control studies quantify how communication constrains stability and performance [26]–[28]. Modern surveys emphasize that information has several operational meanings in estimation and control [29]. Feedback itself consumes network opportunities and can reverse freshness conclusions [30]. Our break-even analysis is deliberately modest: it asks whether the measured performance headroom can pay for the messages needed to expose a hidden value statistic under a frozen packet-frequency accounting model.

III. Sampled Distributed Formation and Communication Model

A. Formation dynamics

Agent 1 is an analytical leader and $\mathcal{F} = \{2, \dots, N\}$ are followers. At sample k, follower i has position $p_i \in \mathbb{R}^3$, velocity $v_i \in \mathbb{R}^3$, and input u_i. The implemented semi-implicit double-integrator update is

$$v_i(k+1) = v_i(k) + hu_i(k), \tag{1}$$
$$p_i(k+1) = p_i(k) + hv_i(k) + h^2u_i(k). \tag{2}$$

For desired offset r_i, define

$$e_i = p_i - p_1 - r_i, \qquad w_i = v_i - v_1. \tag{3}$$

The directed convention is $A_{ij} = 1$ when controller i uses a packet sent by j. The evaluated graph is fixed and its follower-grounded dynamics are Schur under the declared gains.

TABLE I
CLOSEST RESEARCH THREADS AND THE DISTINCTION OF THIS PAPER.

Thread	Established capability	Remaining question	Position of this paper
Event-triggered and lossy formation	Stability/performance triggers under distributed state error, loss, or delay	Is exact marginal action value available to the transmitter?	No new trigger; a valid bound-trigger counterexample motivates the information question.
AoI / goal-oriented freshness	Timeliness and incorrectness metrics	Does freshness determine the closed-loop action cross term?	AoI is not treated as a sufficient control-value statistic.
VoI scheduling	Marginal control benefit and threshold scheduling under specified information	Is the value statistic itself identifiable under a distributed sender map?	Studies this precondition; makes no optimal-scheduler claim.
Decentralized information structures	Formal dependence of decisions on distributed histories	Which exact functional is missing for communication decisions?	Instantiates the information question with finite-horizon communication-action value.
Functional and sample-based observability	Row-space/modal/structural tests, sampling conditions, observer design, and target-sensor selection	What irreducible value and decision loss follows when a communication-action functional is hidden?	Uses established tests; adds action-specific minimax regret, simultaneous-action dimension, acquisition separation, and formation closure.

Receiver i stores the newest accepted ordinary neighbor payload $\hat{x}_{ij} = (\hat{p}_{ij}, \hat{v}_{ij})$. Pinned followers separately store $(\hat{p}^L_{i1}, \hat{v}^L_{i1}, \hat{a}^L_{i1})$. Newer-generation acceptance prevents an obsolete reordered packet from replacing newer memory. DATA may be lost or delayed; an ACK architecture may separately report acceptance, but the information-limit theorem only uses the variables actually supplied to the sender.

Before acceleration saturation, the implemented control law is

$$\begin{aligned} u_i = & -K_p s_i \sum_j A_{ij}\left[(p_i - \hat{p}_{ij}) - (r_i - r_j)\right] \\ & - K_v s_i \sum_j A_{ij}(v_i - \hat{v}_{ij}) - \pi_i K_{pL}[(p_i - \hat{p}^L_{i1}) - r_i] \\ & - \pi_i K_{vL}(v_i - \hat{v}^L_{i1}) + \pi_i \hat{a}^L_{i1}. \end{aligned} \tag{4}$$

Its exact-information part is $u^\star_F = -H_p e - H_v w + \Pi \mathbf{1} a_1$; stale memories add a command residual d^c. With $y = \mathrm{col}(e, hw)$, the exact unsaturated sampled update can be written

$$y_{k+1} = A_h y_k + D_h \,\mathrm{col}(\chi_k, \upsilon_k), \tag{5}$$

where χ_k is the analytical-leader position-step residual and $\upsilon_k = h^2 d^c_k + h^2 \Pi \mathbf{1} a_1(k) - h\mathbf{1}[v_1(k+1) - v_1(k)]$. This form follows from the actual velocity-first integrator; it is not a continuous-time approximation.

B. Information available at a sender

At a decision by source j, the strongest deterministic local information used in this paper contains its exact own position and velocity; every ordinary and pinned receiver memory feeding its own controller; its exact unsaturated command; graph, gains, offsets, clock, and channel law; and its own sent-DATA history. The leader additionally knows its reference position, velocity, and acceleration. It does not include state stored at another receiver or remote physical state that has not been communicated. We express the state-dependent part as

$$o_j = C_j \xi + d_j. \tag{6}$$

Known packet histories or action labels change affine constants but do not add hidden-state rows. Figure 1 illustrates the distinction.

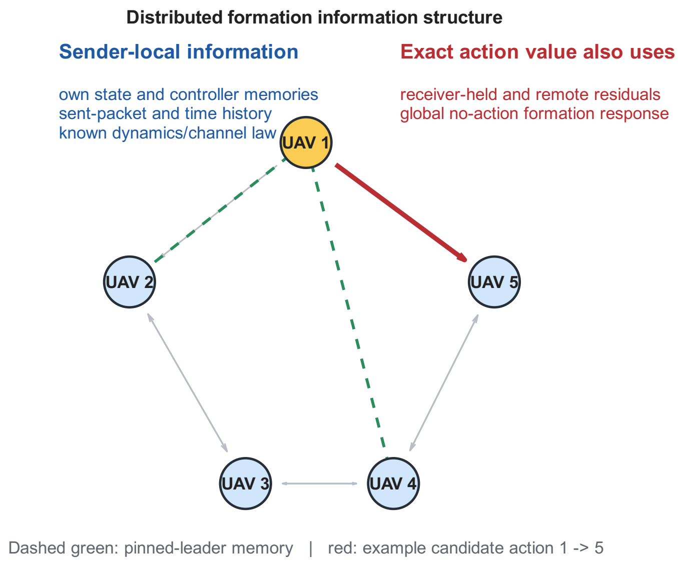

Fig. 1. Evaluated distributed information architecture. The exact value of the highlighted candidate depends on the global no-action response, not only on the sender's own state and estimate of receiver memory.

IV. FINITE-HORIZON VALUE OF A COMMUNICATION ACTION

A. Generic linear/affine closed loop

Consider a finite-dimensional affine closed loop

$$x_{k+1} = Ax_k + Bu_k + d. \tag{7}$$

For a fixed decision point, collect the no-action performance output over a finite horizon as

$$z = Fx + r. \tag{8}$$

A candidate communication action a induces a fixed response g_a in the same stacked output. With symmetric $Q \succeq 0$, define benefit as no-action cost minus action cost:

$$\begin{aligned} q_a(x) &= \|Fx + r\|^2_Q - \|Fx + r + g_a\|^2_Q \\ &= \ell_a^\top x + c_a, \end{aligned} \tag{9}$$

where

$$\ell_a = -2F^\top Q g_a, \qquad c_a = -2r^\top Q g_a - g_a^\top Q g_a. \tag{10}$$

Positive normalizations, including a delivery probability, do not change identifiability or sign. If the response depends on an unknown state, that state must be augmented or the response must be conditioned on a hypothesis; otherwise (9) does not apply.

B. Formation action response

For the formation case, the augmented state contains exactly the variables needed to construct the frozen no-action response: follower formation error and scaled relative velocity, analytical-leader state, eight ordinary receiver memories, and two pinned-leader memories. For one Cartesian axis,

$$\begin{aligned}\xi^a = \operatorname{col}\big(&e^a, hw^a, p_L^a, hv_L^a, h^2 a_L^a,\\ &\{\hat{p}_{ij}^a, h\hat{v}_{ij}^a\},\\ &\{\hat{p}_{iL}^a, h\hat{v}_{iL}^a, h^2\hat{a}_{iL}^a\}\big).\end{aligned} \tag{11}$$

The one-axis dimension is 33 and $\xi = \operatorname{col}(\xi^x, \xi^y, \xi^z) \in \mathbb{R}^{99}$.

Recursion of (5) over H samples yields implementation-derived matrices, not fitted matrices,

$$\operatorname{vec}(Z_k) = F_H \xi_k + r_H. \tag{12}$$

For link $j \to i$, receiver-memory hypothesis s, successful-delivery probability p_s, $m = N - 1$, and fixed output response $g_{ij,s}$,

$$q_{ij,s} = -\frac{2p_s}{m} g_{ij,s}^\top (F_H \xi_k + r_H) - \frac{p_s}{m} \|g_{ij,s}\|_2^2. \tag{13}$$

For a receiver-memory belief $b(s)$, the exact conditional expression is $\bar{q}_{ij} = \sum_s b(s) q_{ij,s}$. The coefficient and constant are averaged across hypotheses. We do not evaluate the nonlinear quadratic at an expected receiver state.

V. Information-Structure Identifiability

A. Exact current value

The following is the action-value specialization of functional observability.

Theorem 1 (Exact action-value identifiability)**.** *Let $q_a(x) = \ell_a^\top x + c_a$ and $o = Cx + d_o$. On $\mathbb{R}^n$, or locally on a feasible set containing an open relative neighborhood along* $\ker C$*, the exact action value is a function of o if and only if*

$$\ell_a \in \operatorname{row}(C). \tag{14}$$

Proof. If $\ell_a = C^\top \alpha$, then $q_a = \alpha^\top (o - d_o) + c_a$. Conversely, any $v \in \ker C$ leaves o unchanged. Identifiability requires $q_a(x + v) - q_a(x) = \ell_a^\top v = 0$ for every such v. Hence $\ell_a \in (\ker C)^\perp = \operatorname{row}(C)$. □

The normalized diagnostic used later is

$$\delta_C(\ell) = \frac{\|(I - C^\dagger C)\ell\|_2}{\max(\|\ell\|_2, \epsilon)}. \tag{15}$$

A positive numerical residual applies the theorem to a given matrix; it is not itself the theorem.

B. Finite local history

Suppose that during a declared interval the augmented dynamics and measurements are

$$x_{t+1} = A_0 x_t + a_0, \qquad o_t = Cx_t + d. \tag{16}$$

Define

$$O_L = \operatorname{col}(C, CA_0, \ldots, CA_0^L). \tag{17}$$

Theorem 2 (Finite-history action-value identifiability)**.** *At time L, an action value with coefficient ℓ_a is identifiable from $o_0, \ldots, o_L$ if and only if*

$$(A_0^L)^\top \ell_a \in \operatorname{row}(O_L). \tag{18}$$

After the observability row space has stabilized, later samples add no observation direction. They cannot recover a decision-time value whenever its propagated coefficient remains outside that stabilized row space.

Proof. The affine solution gives $x_L = A_0^L x_0 + \sum_{t=0}^{L-1} A_0^{L-1-t} a_0$. Thus the coefficient of the time-L value in x_0 is $(A_0^L)^\top \ell_a$, while the stacked history's state-dependent part is $O_L x_0$. Theorem 1 gives (18). Cayley–Hamilton implies that the row span generated by $C, CA_0, \ldots$ stabilizes in finite dimension. □

The last sentence is conditional on unchanged maps and a coefficient that retains its hidden component. A hidden mode annihilated by the dynamics can cease to affect a later value; that is loss of dependence, not recovery of information. A delivered packet, new sensor, or shared statistic changes the information structure.

C. Irreducible value and decision uncertainty

Exact magnitude is stronger than the sign needed by a transmit-if-beneficial rule. Let $\mathcal{X}$ be the declared operating set,

$$\mathcal{X}(o) = \{x \in \mathcal{X} : Cx + d_o = o\}, \tag{19}$$

and assume its image under q is the finite interval

$$Q(o) = [q_-(o), q_+(o)]. \tag{20}$$

Define the midpoint and value-information radius

$$q_c(o) = \frac{q_+(o) + q_-(o)}{2}, \qquad \Delta_q(o) = \frac{q_+(o) - q_-(o)}{2}. \tag{21}$$

Theorem 3 (Irreducible value and communication-decision uncertainty)**.** *At a fixed information state o:*

1) *the minimax error of any sender-local point estimate is*

$$\inf_{\widehat{q}(o)} \sup_{x \in \mathcal{X}(o)} |\widehat{q}(o) - q(x)| = \Delta_q(o), \tag{22}$$

and the midpoint $\widehat{q} = q_c$ is optimal;

2) *if $q_- < 0 < q_+$, no deterministic map of o recovers the exact value sign for every compatible state;*
3) *for a binary decision whose exact-information rule transmits iff $q > 0$, a sender-local randomized rule transmitting with probability p has local endpoint regrets*

$$R_+(p) = (1 - p) q_+, qquad R_-(p) = p(-q_-). \tag{23}$$

Its minimax probability and unavoidable regret are

$$p^* = \frac{q_+}{q_+ + (-q_-)}, \qquad R^*_{\text{rand}} = \frac{q_+(-q_-)}{q_+ + (-q_-)} > 0. \quad (24)$$

Restricting to deterministic rules gives

$$R^*_{\text{det}} = \min\{q_+, -q_-\} > 0. \quad (25)$$

Proof. For any scalar estimate y, $q_+ - q_- \le |q_+ - y| + |y - q_-|$, so its worst endpoint error is at least $(q_+ - q_-)/2$; the midpoint attains that bound throughout the interval. If the interval straddles zero, the same observation is compatible with opposite exact signs, proving the deterministic statement. For the randomized action, regret over positive values is maximized at q_+ and decreases with p; regret over negative values is maximized at q_- and increases with p. Equating $(1-p)q_+ = p(-q_-)$ yields (24). The deterministic choices $p = 0$ and $p = 1$ incur worst regrets q_+ and $-q_-$, respectively, proving (25). □

For $q(x) = a^\top x + b$, let $P_C = C^\dagger C$, $a_\perp = (I - P_C)a$, and consider the Euclidean local fiber $x = x_0 + v$, $v \in \ker C$, $\|v\|_2 \le \rho$. Its exact interval is

$$\begin{aligned} Q(o) = [&q(x_0) - \rho\|a_\perp\|_2, \\ &q(x_0) + \rho\|a_\perp\|_2], \\ \Delta_q(o) = \rho&\|a_\perp\|_2. \end{aligned} \quad (26)$$

The endpoints follow by choosing v parallel or antiparallel to $a_\perp$. Figure 5 shows the geometry.

Theorem 3 is a local minimax statement over $\mathcal{X}(o)$, not a universal stochastic-control lower bound. It does not rule out a Bayesian conditional expectation under a specified prior, nor does it say that every information fiber crosses zero.

VI. Minimum Supplementary Information and Coalition Structure

A. One or multiple candidate actions

Stack p action-value coefficients as rows of $L \in \mathbb{R}^{p\times n}$ and define

$$L_\perp = L(I - C^\dagger C). \quad (27)$$

Theorem 4 (Minimum linear-statistic dimension)**.** *The minimum number of additional linear scalar statistics Hx required to identify all p exact action values is*

$$r_\star = \operatorname{rank}(L_\perp). \quad (28)$$

Proof. If all values are identifiable from $\operatorname{col}(C, H)x$, the projections of the rows of H onto $\ker C$ must span every row of $L_\perp$. Therefore H has at least $\operatorname{rank}(L_\perp)$ independent scalar rows. Conversely, choose the rows of H as any row basis of $L_\perp$. Each row of L is then the sum of a component in $\operatorname{row}(C)$ and one in $\operatorname{row}(H)$, satisfying Theorem 1 for all p values. □

For a single nonidentifiable value, one additional scalar $\theta = \ell_\perp^\top x$ is sufficient. This is statistic dimension, not packet count. The coordinates of $\ell_\perp$ may be distributed across many agents.

Corollary 1 (Instantaneous linear supplementary messages)**.** *If the decision maker can receive r independent real-valued linear scalar statistics before deciding, exact identification of all p values requires $r \ge r_\star$. Equality is algebraically achievable if the received statistics span* $\operatorname{row}(L_\perp)$.

This is a message-dimension statement, not a data-rate theorem. Scalar dimension, packet count, bits, latency, functional-observer order, and the distributed cost of synthesizing each scalar are distinct quantities.

B. Information ownership

Let agent i's strongest map be C_i. For coalition $\mathcal{S}$, define $C_{\mathcal{S}} = \operatorname{col}_{i\in\mathcal{S}} C_i$.

Corollary 2 (Coalition identifiability)**.** *Coalition $\mathcal{S}$ identifies a fixed action value if and only if*

$$\ell \in \operatorname{row}(C_{\mathcal{S}}). \quad (29)$$

A minimum owning coalition solves

$$\min_{\mathcal{S}} |\mathcal{S}| \quad \textit{subject to} \quad \ell \in \operatorname{row}(C_{\mathcal{S}}). \quad (30)$$

Corollary 2 is closely related to functional sensor selection. We use exhaustive enumeration only for the N=5 case and do not claim a new generic combinatorial algorithm. Even when a coalition owns the required rows, a causal aggregation protocol must still communicate them at a nonzero frequency, delay, and reliability cost.

VII. Communication Cost of Acquiring Value Information

Let $C_P(E)$ be the cost of the frozen periodic Pareto frontier interpolated at formation error E. Suppose a policy informed by supplementary statistics uses DATA cost D and information streams t with rate R_t and normalized price η_t. A necessary accounting condition for performance-matched Pareto benefit is

$$D + \sum_t \eta_t R_t < C_P(E). \quad (31)$$

For a single feedback stream with rate A,

$$\eta^\star = \frac{C_P(E) - D}{A} \quad (32)$$

is its break-even price. Negative $\eta^\star$ means DATA alone is already more costly than periodic at matched error. At feedback price η_f, the affordable feedback fraction is $f^\star = \eta^\star/\eta_f$, clipped to $[0, 1]$ for operational interpretation.

Equation (31) is not an optimality theorem. It prevents a common category error: a one-dimensional sufficient statistic is not free. Under a packet-frequency metric, piggybacking adds no packet only if an existing packet is sent anyway; bytes, airtime, collision risk, and joules remain outside that metric.

VIII. Formation-Control Case Study

A. Implementation-derived matrices and validation

The case study uses N=5, $h = 0.02$ s, four followers, eight ordinary controller links, two pinned-leader streams, horizon H=25, and DATA delay D=4 samples. The augmented state has 33 entries per axis and 99 in 3-D. The matrices F_H, action responses, sender maps, and held-memory dynamics

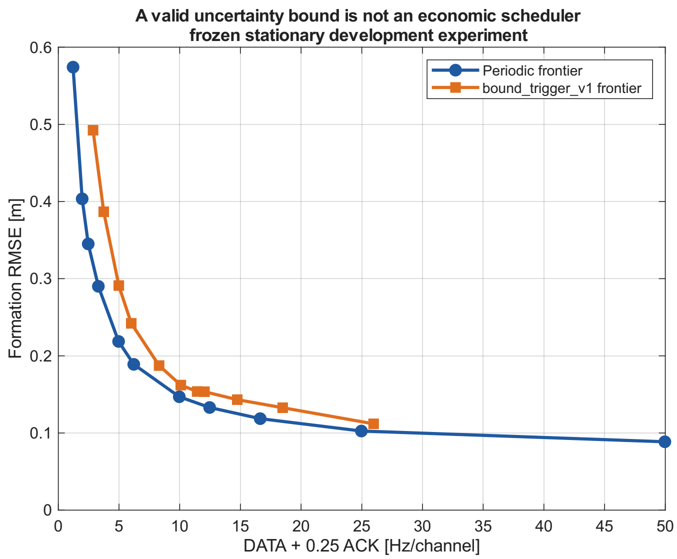

Fig. 2. Frozen stationary development frontiers. The analytically valid bound-based mechanism is dominated by the periodic frontier over their shared domain.

are assembled from the implemented graph, gains, controller, and integrator. Across all ten payloads, the maximum discrepancy between the affine representation and the centralized O1 computation is 4.34×10^{-16}.

For a fixed receiver-memory hypothesis, its coordinates are substituted from the free state. Reduced dimensions are 93 for ordinary actions and 90 for pinned-leader actions. The history analysis uses L=92 and L=89 respectively, which reaches the complete finite-dimensional held-memory observability horizon.

Table II shows that all ten value coefficients lie outside the strongest sender-history row spaces. The small residuals for leader payloads are not treated as negligible: their missing projections are nonzero, and the dynamic witness below makes their decision consequence explicit. Adding the single missing scalar closes the row-space test for every action.

The nonidentifiability decision is unchanged for relative tolerances 10^{-6} through 10^{-14}. Retained/discarded gaps span 3.08×10^{11} to 1.32×10^{12}, and appending the value row increases rank by one for all ten actions. At 10^{-14}, three follower maps retain two roundoff-level modes and their raw rank changes; even then their normalized value residuals remain 0.071–0.080 and the conclusion does not change. Leader and pinned-payload ranks are stable across the full grid. An independent 80-digit row-basis projection reproduces every double residual within relative difference 2.4×10^{-15}, including the smallest 0.003626 value. The high-precision audit is conditional on the implementation-generated matrices; it is not a symbolic arbitrary-gain proof.

B. Reachable value-sign witnesses

Affine nonidentifiability alone could be dismissed as an inadmissible state perturbation. We therefore preselected one ordinary leader payload $1 \to 5$ and one pinned-leader payload $1 \to 4$. The same small constant-velocity leader trajectory, sent packet, positive-probability no-delivery history, receiver-memory belief, and action response are imposed in each pair. Follower initial conditions differ only along a null-space direction of the complete leader information history. The existing formation controller and integrator then generate both full trajectories.

For the ordinary payload, the belief-weighted exact values are -5.29704×10^{-6} and $+5.29704 \times 10^{-6}$; the values using the realized receiver memory are -5.33798×10^{-6} and $+5.34903 \times 10^{-6}$. For the pinned payload, the corresponding belief-weighted values are $\mp 4.95740 \times 10^{-6}$. The maximum sender observation and action-response differences are zero to stored precision. The affine trajectory residual is below 3×10^{-16}, and the maximum unsaturated command is below 0.065 m/s^2 against a 2 m/s^2 limit.

Applying Theorem 3 to the compatible expected-value endpoints gives the quantitative limits in Table III. Both intervals are nearly symmetric, so p^* is one half, randomized minimax regret is half the value-information radius, and the best deterministic rule incurs the full radius. These are local losses at the constructed information states, not expected mission-level or universal stochastic-control bounds.

Figure 6 shows the two dynamically distinct but sender-indistinguishable histories. Theorem 3 therefore applies at these two reachable information states. We claim reachable sign ambiguity for these two payload classes; Table II supplies affine nonidentifiability, not dynamic sign witnesses, for the remaining eight.

C. Statistic dimension versus coalition size

For every fixed payload, $\mathrm{rank}(L_\perp) = 1$: one linear scalar closes its individual algebraic deficit. However, no receiver alone and no sender–receiver pair identifies that scalar. Exhaustive enumeration of all subsets of the five implemented agent maps finds that every sender needs four additional agents; the only minimum coalition contains all five UAVs. This is a result for the evaluated graph and information topology, not a claim that all formations require global information.

The fixed-action statement underuses Theorem 4. We therefore stack every controller-relevant current action of each actual sender in L_j, using a common full 99-dimensional state and the structural unit command response. Table IV reports the simultaneous missing rank. Sender 1 has four payloads but $r_1^* = 3$: its ordinary and pinned-leader unit actions into follower 2 induce the same output-response direction, so one hidden direction is shared. Senders 3 and 4 each require two independent directions for two actions; the single-action senders require one. Thus compression exists for the leader but is not universal. The nonzero singular values shown in the table are those of $L_j(I - C_j^\dagger C_j)$.

Although $r_1^* = 3 < 4$, the minimum coalition for its entire missing subspace still contains all five agents. The same coalition size holds for every other sender's full action set. Compression of value directions therefore does not imply local ownership or three packets: Corollary 1 counts independent real-valued statistics only if their operands can be assembled before the decision.

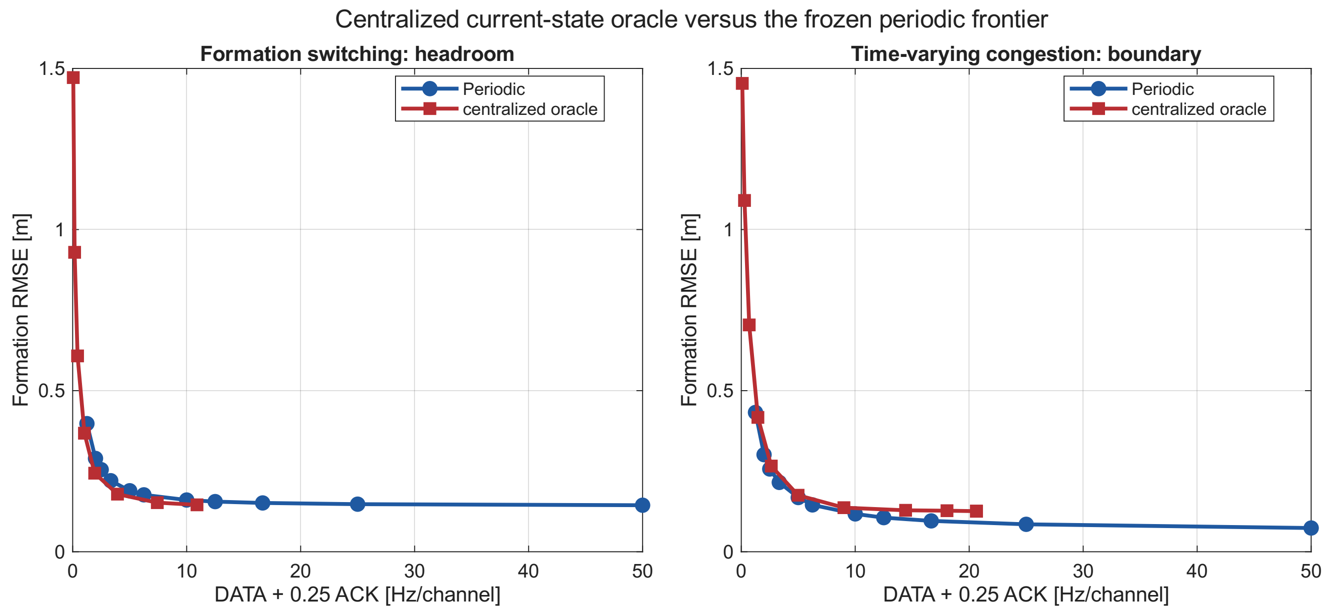

Fig. 3. Frozen centralized-current-information diagnostic. Exact current value creates Pareto headroom in S2 but not in the S4 congestion boundary, demonstrating both usefulness and non-universality.

TABLE II
N=5 FINITE-HISTORY ACTION-VALUE AND RANK AUDIT. ORDINARY/PINNED HISTORIES USE L=92/89. HERE R IS THE REFERENCE INFORMATION RANK, AND σ_r, σ_{r+1} BRACKET THE RETAINED/DISCARDED SINGULAR-VALUE GAP. THE LAST COLUMNS COMPARE DOUBLE AND 80-DIGIT NORMALIZED PROJECTION RESIDUALS.

Payload	Link	n	r	σ_r	σ_{r+1}	δ_{dbl}	δ_{80}
ordinary	$1 \to 2$	93	9	3.288	6.86×10^{-12}	0.056035	0.056035
ordinary	$1 \to 5$	93	9	3.288	6.86×10^{-12}	0.010226	0.010226
ordinary	$2 \to 3$	93	27	2.776	8.99×10^{-12}	0.999890	0.999890
ordinary	$3 \to 2$	93	18	3.150	2.38×10^{-12}	0.999921	0.999921
ordinary	$3 \to 4$	93	18	3.150	2.38×10^{-12}	0.999922	0.999922
ordinary	$4 \to 3$	93	27	2.776	9.01×10^{-12}	0.999889	0.999889
ordinary	$4 \to 5$	93	27	2.776	9.01×10^{-12}	0.999876	0.999876
ordinary	$5 \to 4$	93	18	3.150	2.38×10^{-12}	0.999922	0.999922
pinned leader	$1 \to 2$	90	9	3.237	7.02×10^{-12}	0.004119	0.004119
pinned leader	$1 \to 4$	90	9	3.237	7.02×10^{-12}	0.003626	0.003626

TABLE III
REACHABLE LOCAL VALUE UNCERTAINTY AND DECISION REGRET.

Payload	Δ_q	p^*	R^*_{rand}	R^*_{det}
ordinary $1 \to 5$	5.2970×10^{-6}	0.5000	2.6485×10^{-6}	5.2970×10^{-6}
pin $1 \to 4$	4.9574×10^{-6}	0.5000	2.4787×10^{-6}	4.9574×10^{-6}

D. Why the information result matters operationally

Figure 2 retains the mechanism that failed before this theory was formulated. Its trigger uses a valid staleness-to-control uncertainty bound, yet the complete periodic frontier is better throughout the shared development domain. Across a 101-point budget-matched grid, trigger-minus- periodic RMSE averages +0.03023 m and no point favors the trigger. This is a counterexample to the inference that a safe uncertainty bound is automatically a useful scheduler; it is not a population comparison.

To test whether valuable scheduling opportunities existed at all, a centralized online oracle used exact current plant and receiver memories but no future channel outcomes, disturbances, formation changes, or trajectories. It ranked actions only by (13). Figure 3 shows a positive and a negative boundary. In S2 formation switching, oracle minus periodic mean matched RMSE is -0.02381 m and mean matched cost is -1.7088 Hz/channel; the oracle is favored over the complete shared domains. In S4 time-varying congestion, the differences reverse to +0.02595 m and +0.7135 Hz/channel, with zero shared-domain fraction favoring the oracle. The oracle is privileged and is not a proposed algorithm.

Finally, Fig. 4 charges the information channel using (32). At the frozen ACK price 0.25, S2 can fund complete feedback at all four matchable points, while the median S6 point can fund feedback on only 37.8% of eligible deliveries. S4 has negative headroom even before feedback. Table V reports all five scenarios. This accounting explains why mathematical sufficiency and deployable economic value must remain distinct.

TABLE IV
SENDER-WISE SIMULTANEOUS-ACTION MISSING DIMENSION AND OWNERSHIP.

Sender j	p_j	r_j^*	r_j^*/p_j	Retained singular values of $L_j^\perp$	Minimum coalition	Interpretation
1	4	3	0.75	26.9758; 20.5237; 1.83164	$\{1, 2, 3, 4, 5\}$	one shared direction
2	1	1	1.00	49.7427	$\{1, 2, 3, 4, 5\}$	single action
3	2	2	1.00	55.0690; 16.5170	$\{1, 2, 3, 4, 5\}$	independent directions
4	2	2	1.00	61.7396; 5.34031	$\{1, 2, 3, 4, 5\}$	independent directions
5	1	1	1.00	45.7777	$\{1, 2, 3, 4, 5\}$	single action

TABLE V
FROZEN FEEDBACK ECONOMICS. STATISTICS COVER ONLY O1 POINTS WITHIN THE MEASURED PERIODIC PERFORMANCE DOMAIN.

Item	Statistic	Result	Interpretation
S2 formation switching	eligible 4/14; median $\eta^\star$	0.726 (IQR 0.556–2.186)	Full feedback at price 0.25 is affordable for every eligible point.
S3 burst loss	eligible 11/14; median $\eta^\star$	0.220 (0.186–0.249)	Median point cannot afford full feedback.
S4 congestion	eligible 11/14; median $\eta^\star$	-0.741 ($[-0.814, -0.173]$)	DATA alone is uneconomic throughout the eligible set.
S5 topology perturbation	eligible 11/14; median $\eta^\star$	-0.047 (-0.070–0.173)	Headroom is sparse and not robust to feedback cost.
S6 dynamic excitation	eligible 11/14; median $\eta^\star$	0.095 (0.032–0.367)	Median affordable feedback fraction is 37.8%.

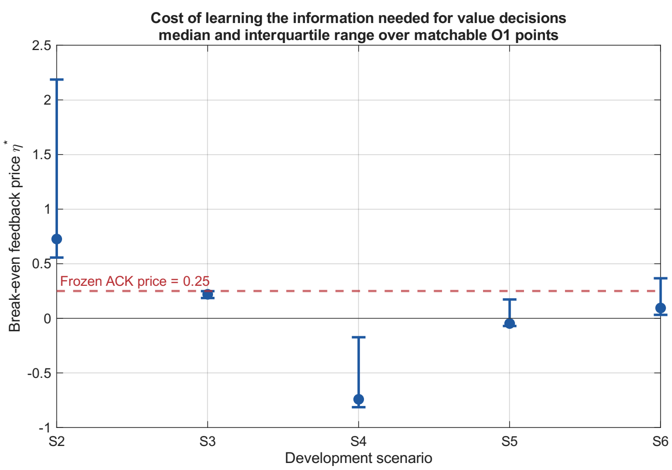

Fig. 4. Break-even feedback price over frozen performance-matchable O1 points (median and IQR). The dashed line is the preregistered price 0.25.

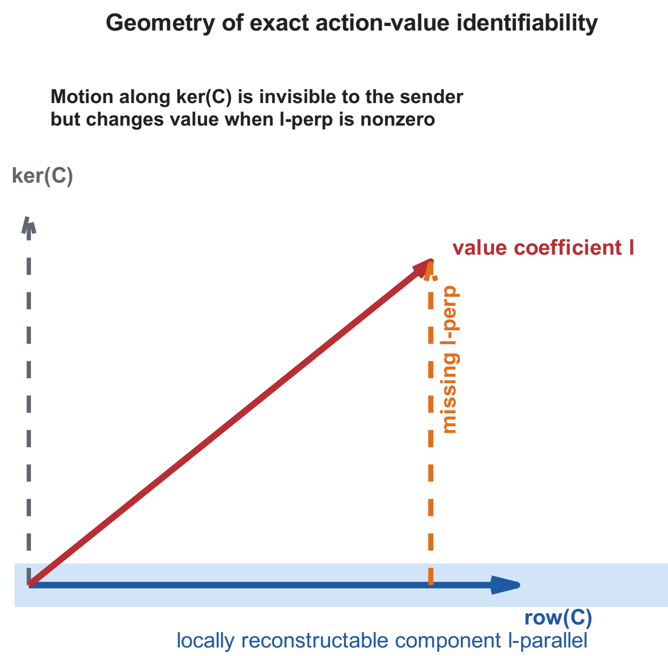

Fig. 5. The available row space determines the reconstructable component of the value coefficient. Motion along the information null space is invisible to the sender but changes value when the missing projection is nonzero.

IX. DISCUSSION AND LIMITATIONS

A. What the result changes

Theorems 1–4 suggest a three-stage audit for control-aware communication. First derive the exact action value under the closed-loop model. Second test whether that functional lies in the acting agent's information row space, including the complete permitted history. Third, if information is missing, distinguish the algebraic dimension of the missing statistic from the coalition and communication resources needed to construct it. Scheduler design before these checks risks optimizing a surrogate that either lacks the decisive cross term or costs more to reveal than its scheduling benefit.

The result also clarifies why receiver-memory estimation alone can be insufficient. Such an estimator can correctly represent which packet the receiver may hold. The finite-horizon action value additionally projects the action response onto the no-action closed-loop response, which includes formation components outside the sender's information. Better calibration of the receiver-memory marginal cannot create measurements of those missing components.

B. Boundaries

The analytical formation case is the unsaturated sampled double-integrator subsystem with a fixed grounded graph. It is not a theorem for nonlinear 6-DOF UAV dynamics, saturation, arbitrary switching topology, or an interference-limited MAC. The N=5 rank and coalition results depend on the implemented graph, gains, and observation maps. Tolerance sweeps and an 80-digit projection audit harden this fixed-configuration conclusion but do not replace a symbolic proof for arbitrary gains or graphs.

The value-radius and regret results are local to an information-compatible operating set. A fully specified stochastic prior and nonlinear Bayesian estimator may produce a useful conditional expected value; that is a different informa-

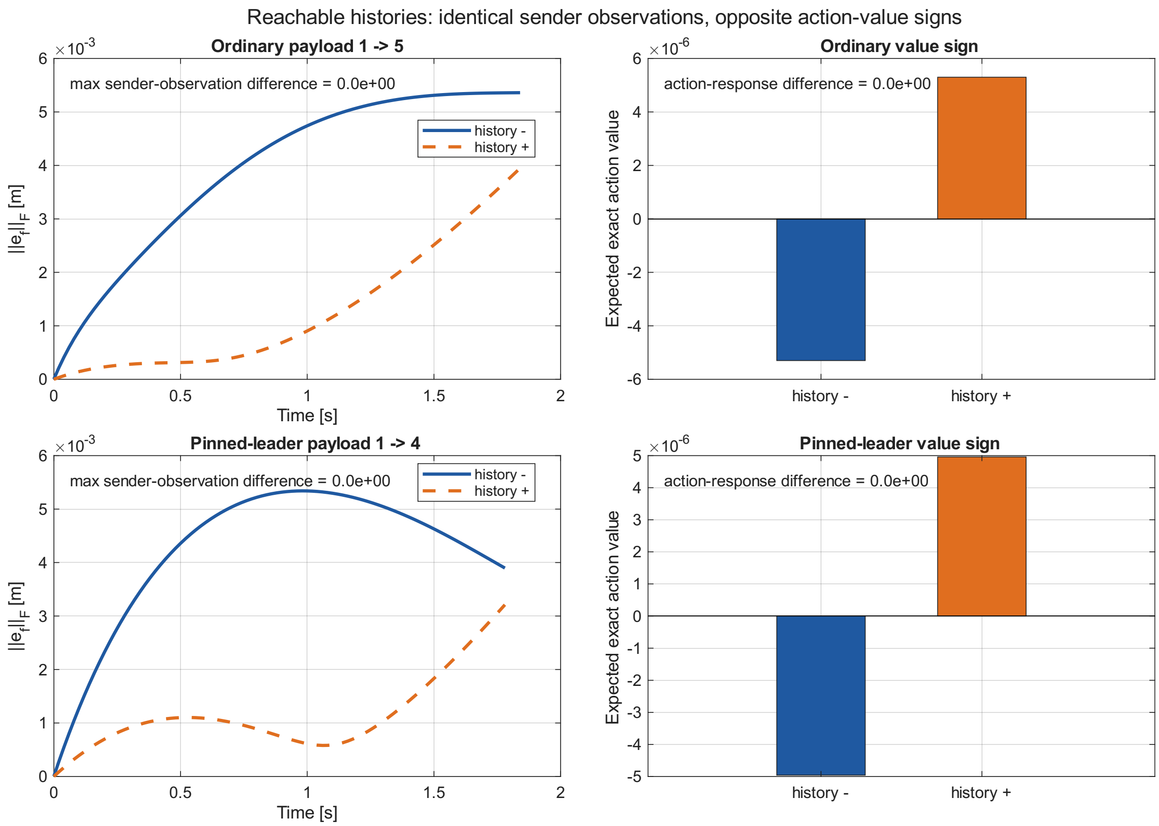

Fig. 6. Full unsaturated formation histories for the two preregistered witnesses. The sender's complete allowed observation history and each candidate action response are identical, although hidden follower evolution and the exact action-value sign differ.

tion model and is not ruled out. Randomization reduces worst-case binary regret for the two nearly symmetric witnesses but cannot remove it or make the hidden exact sign observable.

The frontier evidence uses five paired development seeds and is presented as mechanism falsification and counterexample evidence. No held-out population superiority or statistical significance claim is made. The communication axis counts DATA + 0.25 ACK packets per second per channel. It omits payload length, shared-medium collisions, airtime, and energy, so feedback-economic numbers are conditional on this accounting.

Functional, sample-based, and structural functional observability, functional observer design, and target-sensor selection are established fields. Our exact row-space, finite-history, and coalition conditions must be read as action-value specializations, not as new generic observability or placement theory. The intended contribution is the control-communication synthesis: the exact action functional, irreducible local value/decision loss, shared missing dimensions across simultaneous actions, acquisition cost, and implementation-faithful reachable closure.

X. Conclusion

Control-aware communication is not locally decidable merely because a sender can estimate the receiver's memory. For a fixed communication response in a linear sampled closed loop, quadratic finite-horizon benefit is an affine state functional. Established functional-observability geometry then gives an exact identifiability test; its hidden projection quantifies local minimax value error and strictly positive communication-decision regret when the compatible interval straddles zero. For several action values, the rank of their joint hidden projection gives the minimum instantaneous linear-statistic dimension. The acquisition problem is separate: even a compressed deficit may be distributed across a large coalition and may consume the Pareto headroom it is meant to exploit.

In the evaluated five-UAV formation topology, all ten action values violate sender-history identifiability, and two reachable history pairs require opposite exact decisions while remaining indistinguishable to the sender. The corresponding local randomized minimax regrets are 2.65×10^{-6} and 2.48×10^{-6}; deterministic regrets are twice as large. Four leader actions share three missing directions, while the other multi-action senders have no such compression. The result motivates information-structure and communication-cost audits before a value-based scheduler is engineered. Future work may study structured priors, nonlinear functional observability, and sparse protocols for acquiring the missing statistic, but none is claimed here.

Appendix A
Detailed Formation Affine Map

For one axis, let $T_y\xi = y$ and write the exact held-current communication input as $T_v\xi+t_v$. Initialize $M_0 = T_y$ and $b_0 = 0$.

Then recurse as

$$M_r = A_h M_{r-1} + \begin{bmatrix} I \\ I \end{bmatrix} T_v, \quad (33)$$

$$b_r = A_h b_{r-1} + \begin{bmatrix} I \\ I \end{bmatrix} t_v. \quad (34)$$

With $C_e = [I\ 0]$, stacking $C_e M_r$ and $C_e b_r$ for $r = 1, \ldots, H$ and then for three coordinates yields F_H and r_H in (12). Substitution of a fixed receiver memory removes its coordinates from the free state and moves them into the constant. The action response is generated by the same A_h, delay D, gains, and target row used by the centralized diagnostic.

Appendix B
Euclidean-Fiber Interval Proof

For $v \in \ker C$, $a^\top v = a_\perp^\top v$. Cauchy–Schwarz gives $|a_\perp^\top v| \leq \rho \|a_\perp\|_2$. If $a_\perp \neq 0$, equality is attained at $v = \pm \rho a_\perp / \|a_\perp\|_2$, which lies in $\ker C$. If $a_\perp = 0$, the interval collapses to one point. This proves (26).

Appendix C
Reproducibility Scope

The base theorem validation entry point is `experiments/tcns_information_limits_validation.m`; the R1 depth audit is `experiments/tcns_r1_theory_depth_validation.m`, with unit test `tests/test_tcns_r1_theory_depth.m`. Figure generation is `paper/tcns/manuscript/make_gm_figures.m`. The authoritative GM theorem-validation run, generated from source commit `15177a8`, is `results/tcns_information_limits_validation/2026-09-08_095209`. The authoritative R1 rank/regret/multi-action audit, generated from source commit `05fa200`, is `results/tcns_r1_theory_depth_validation/2026-09-08_103325`. Frozen Gate-5, O1, AB0, and FB0 result directories are named in the repository's GM claim–evidence ledger. Each result records configuration, seed, source commit, MATLAB release, machine-readable tables, and a console log. No held-out seed or new policy experiment is used in this manuscript.

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